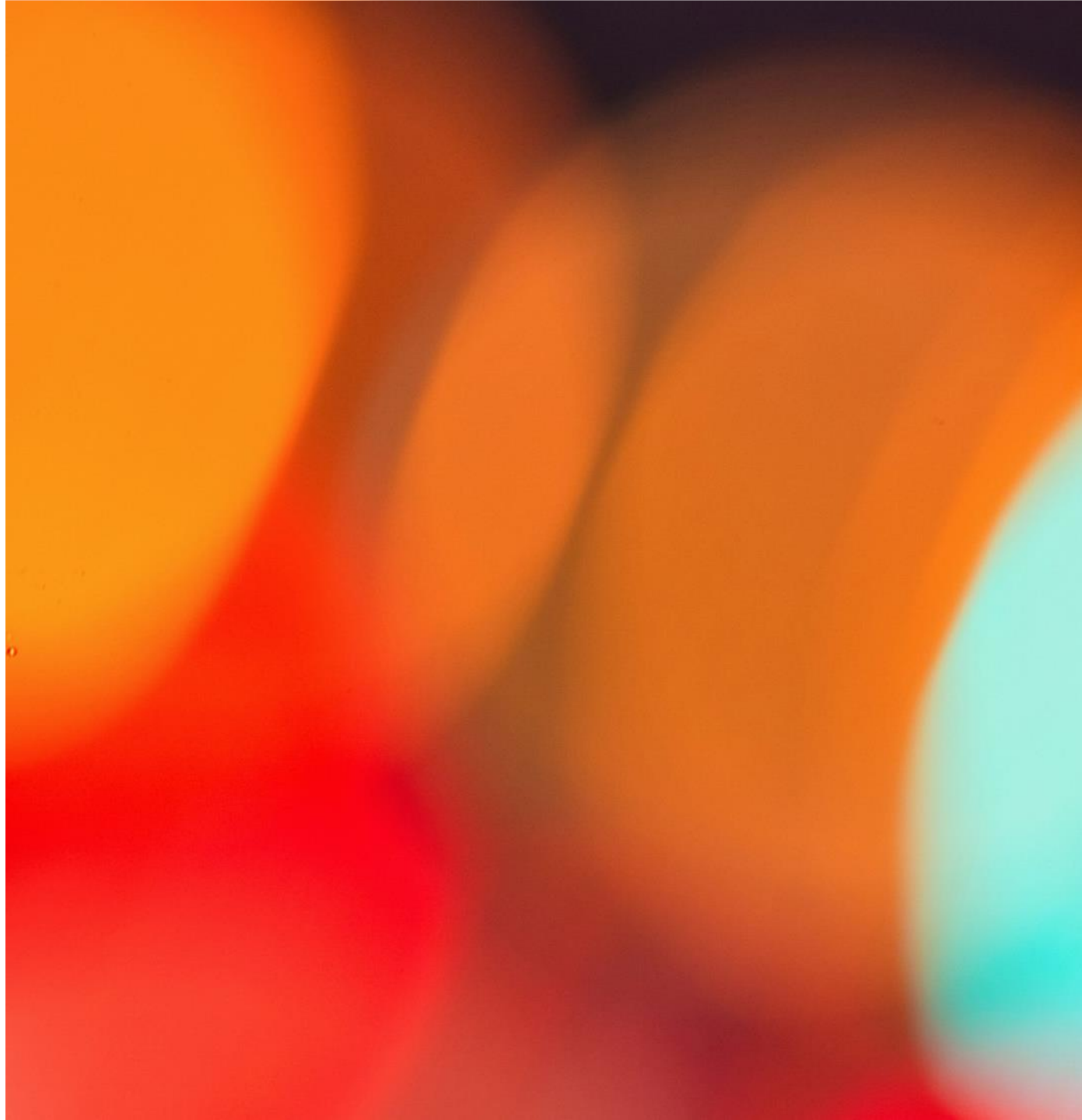
Utility Forge ♪



Contributors: Kevin, Luisa, Pascual, Roman, Vinh

Week 08

Define the user and utility

- Choose one route. Write the story or client brief, intended user, problem, useful result, limits, privacy boundary and assumptions.

1. Open and control a local audio file
2. Store and open reference links
3. Export and reopen session notes

Selected Route

Music Reference Hub

- Build a local audio player with play, pause and stop controls, a reference-link list, and a plain-text session note.

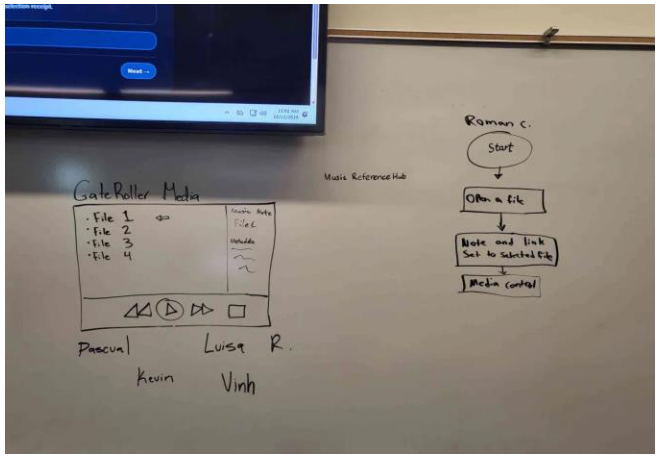
Requirement 01

Story:

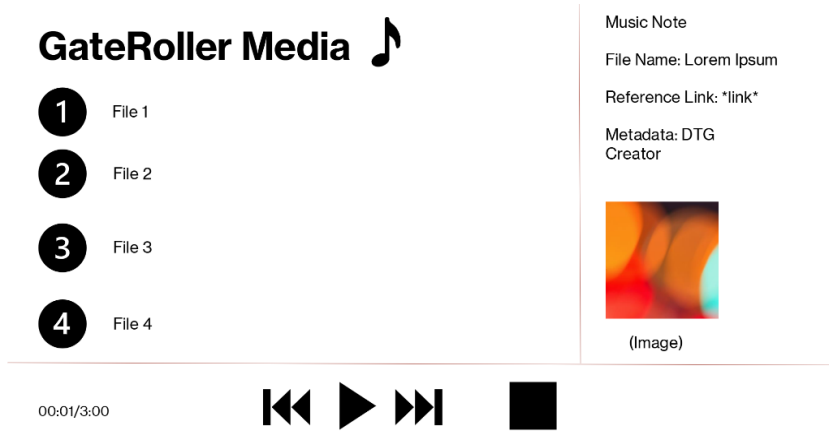
We selected the Music Reference Hub. The intended user is the Gaterollin' class group. The problem is referencing source of music files. The useful result is that you can document the source link with the reference file. There are several limits that are yet to be discovered and worked on. The utility does not send any information online.

Prototype and map the events

- Create the interface prototype and whiteboard logic chart. Label input, event, method, data, output, failure and recovery. Complete one working event.



WB – Prototype + Logic Chart



Prototype

Requirement 02

GateRoller Media

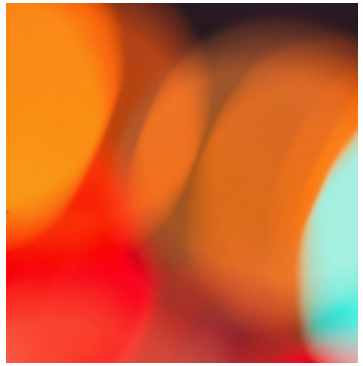
1. File 1
2. File 2
3. File 3
4. File 4

Music Note

File Name: Lorem Ipsum

Reference Link: *link*

Metadata: DTG Creator



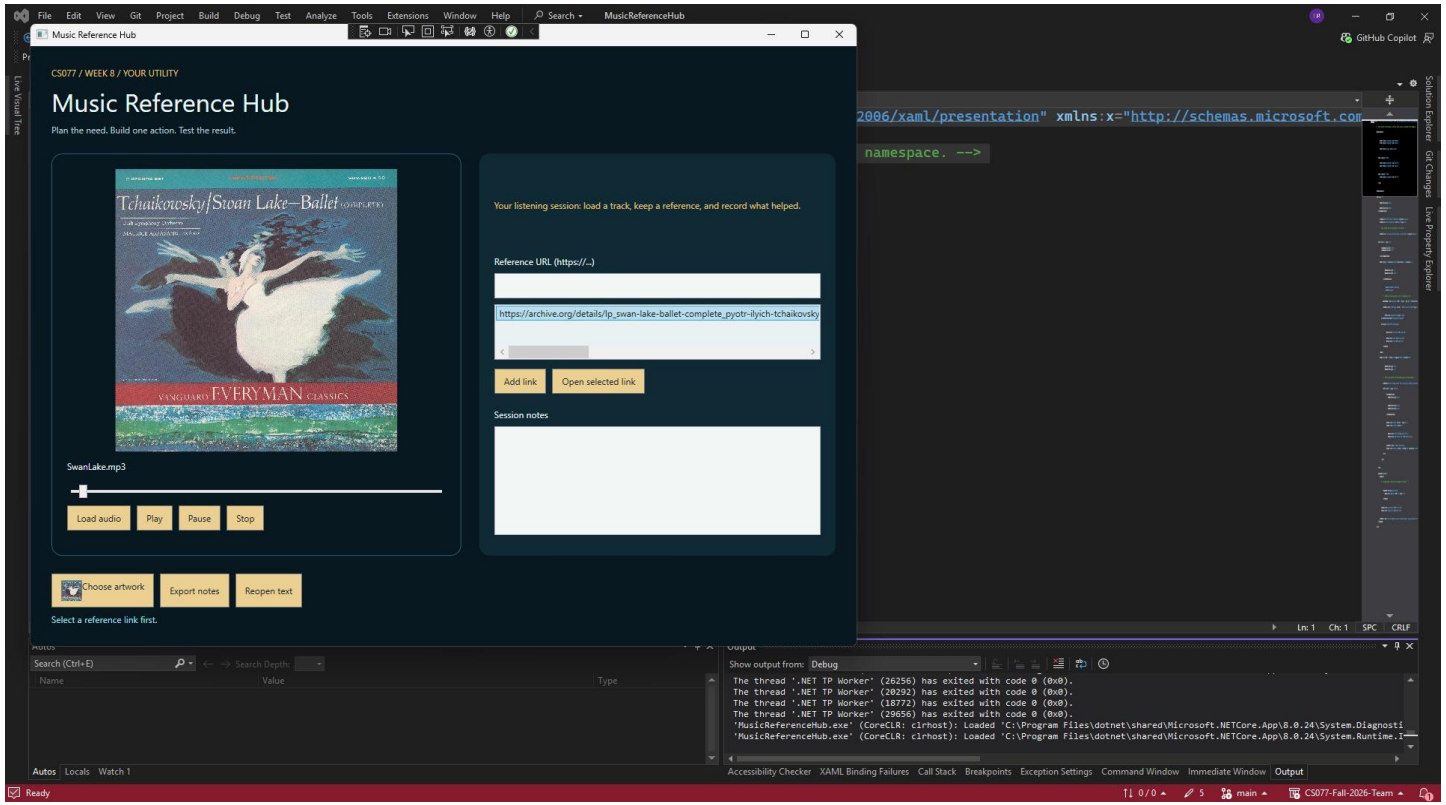
(Image)

00:01/03:00

Prototype

Build the selected utility

- Complete the route's three core functions in one clean WPF app. Use named event handlers, focused methods, a list or collection, and visible status.



Music Reference Hub

Requirement 03

Validate, recover and save text

- Reject blank or unsafe input without crashing. Prove recovery, then export and reopen one plain-text record.

Code with required components

```
13  public partial class MainWindow : Window
    private void OpenLink_Click(object sender, RoutedEventArgs e) => OpenSelectedLink();

    private void SaveText(string text)
    {
        var dialog = new SaveFileDialog { Filter = "Text files|*.txt", FileName = "music-record.txt" };
        if (dialog.ShowDialog() != true)
        {
            txtStatus.Text = "Save canceled. Your text stays here.";
            return;
        }
        try
        {
            File.WriteAllText(dialog.FileName, text);
            txtStatus.Text = "Saved: " + Path.GetFileName(dialog.FileName);
        }
        catch (Exception ex)
        {
            txtStatus.Text = "Could not save. Choose a writable folder: " + ex.Message;
        }
    }

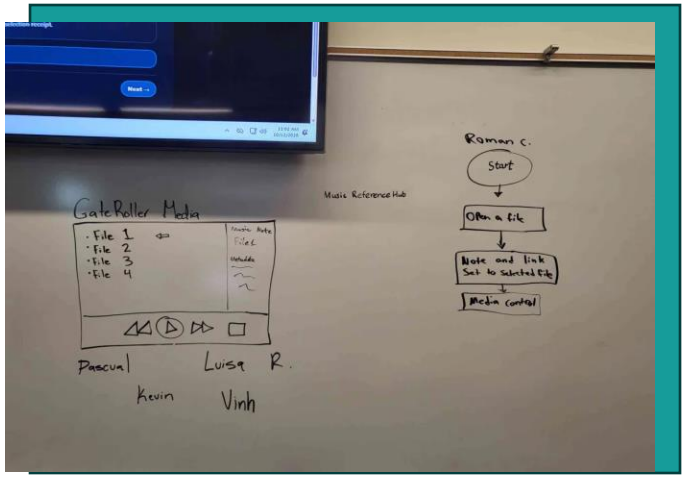
    private void Open_Click(object sender, RoutedEventArgs e)
    {
        var dialog = new OpenFileDialog { Filter = "Text files|*.txt" };
        if (dialog.ShowDialog() != true)
        {
            txtStatus.Text = "Open canceled. Your current text stays here.";
            return;
        }
        try
        {
            txtNotes.Text = File.ReadAllText(dialog.FileName);
            txtStatus.Text = "Text reopened. Review it before editing.";
        }
        catch (Exception ex)
        {
            txtStatus.Text = "Could not read the file: " + ex.Message;
        }
    }
```

Requirement 04

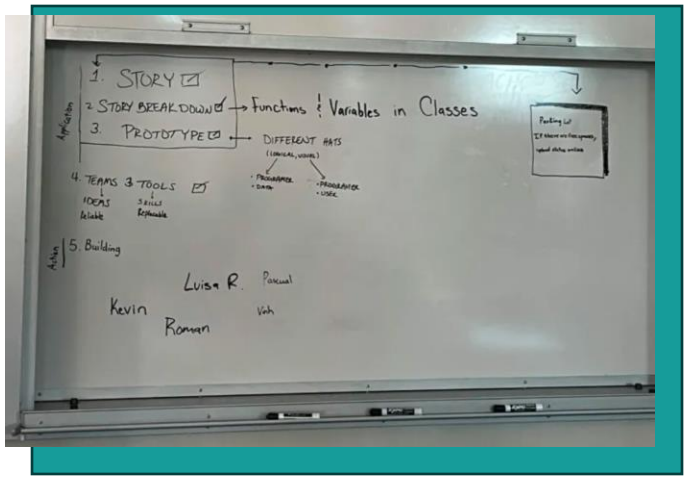
Keep the two class records

- Include 8A and 8B whiteboards, lecture notes, collaboration, concise Discord records, mandatory AI test, screenshots and accepted Git checkpoints. Reuse this evidence in the weekly repository

8A WB/DISCORD:



8B WB/DISCORD:



Requirement 05

Keep the two class records

AI Prompt

- I am building a beginner C# WPF utility. Myselected route is [Operations / Music / Video]. Mycurrent unknown is [ONE precise unknown].Explain the smallest useful change, name the fileand method where it belongs, and give me onetest. Do not rewrite the whole project. I will test theresponse and record what I kept, changed, orrejected.

AI TEST:

```
Focused AI Assistance – Clear Link Selection

Prompt used:
I am building Music Reference Hub in C# WPF. My unknown is when I deselect a link and click Open Selected Link, it still opens the link
again. Explain the smallest useful change, the exact file and method, and one test. Explain the variables and method calls involved. Do
not rewrite my project. I will record what I kept, changed or rejected.

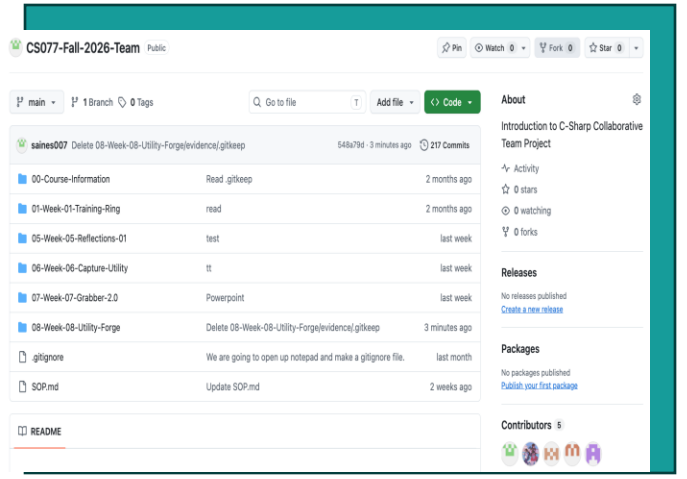
Change:
Added a Window_MouseDown event handler in MainWindow.xaml.cs to clear lstLinks.SelectedItem when clicking outside the links ListBox.
Connected the event handler in MainWindow.xaml using MouseDown="Window_MouseDown".

Decision: KEPT

Reason:
The change fixed the issue. Clicking outside the links box now clears the selection, preventing the previously selected URL from opening
again.

Test:
Input/action: Select a URL, open it, click outside the links ListBox, then click Open Selected Link again.
Predicted result: No browser opens because no link is selected.
Observed result: Worked as expected.
```

GITHUB:



Requirement 05

Document one reusable pattern

- Finish README.md and PATTERN.md. Explain one event, method, validation, UI, timer, media, collection or file pattern that can be adapted later.

README.MD

DTG 141830UOCT26

1. Finished code for action 8A and 8B
2. Code is retained from course tutorial
3. Usage of AI to amplify product
4. Installed TagLibSharp library from NuGet for metadata reading and media control
5. Added Media Control similar to WMP/VLC
6. Added demo video

AI Prompt for Google Gemini: Allow utilization of metadata from music file to display already included track cover image. Set the choose image button function to also be able to replace current cover image display with an image of user choice similar to current code, without permanently overwriting the track's own cover image from metadata.

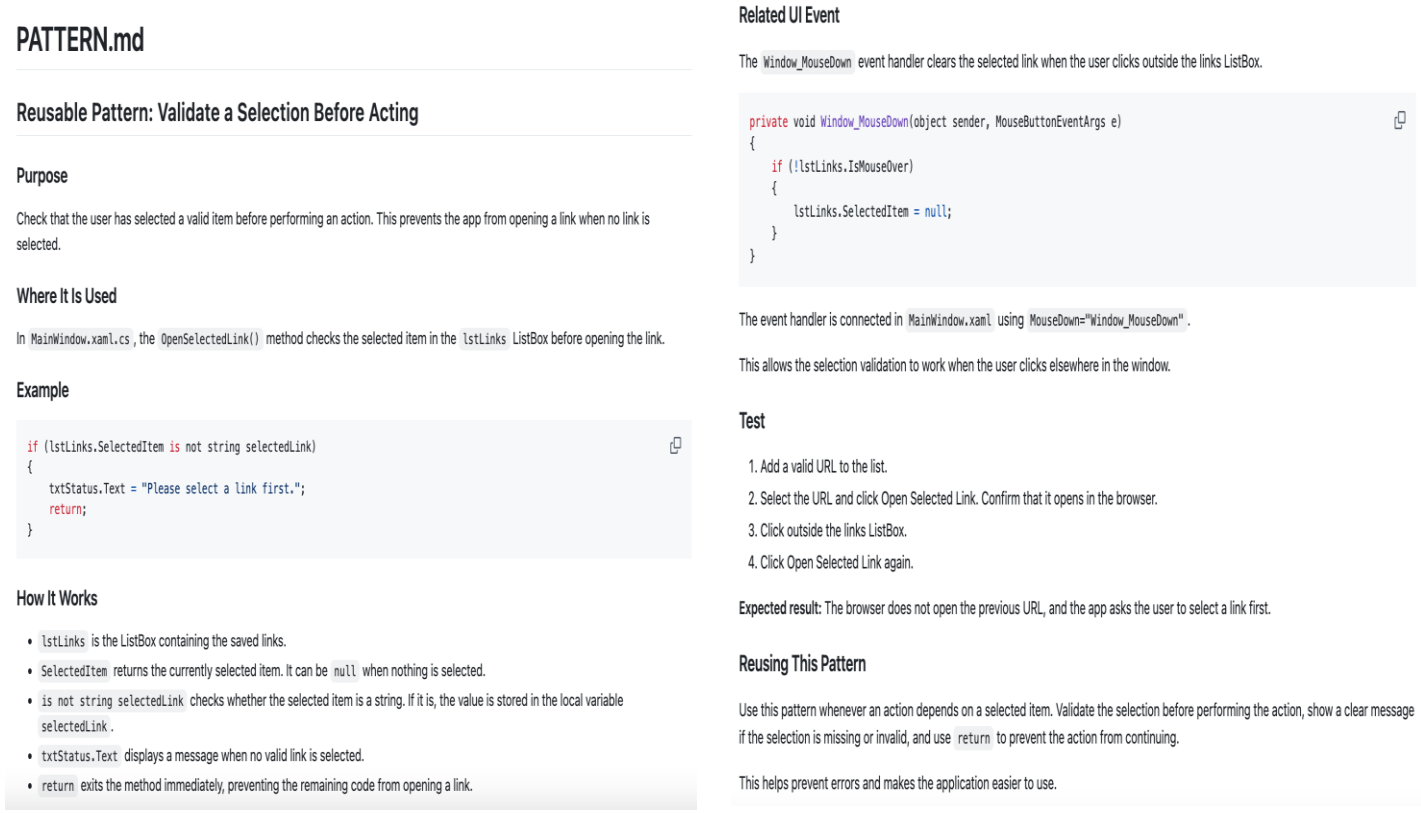
Create progress slider for track similar to youtube/VLC/WMP.

Document one reusable pattern

Explain one event

One of the events we worked on, follows a logical map of events after the user clicks on "Choose Artwork." It opens the file explorer and the user can then select their desired image. After selecting the image, it makes a bitmap out of it. Then it sets it to the imgArtwork.Source and imgButton.Source and when the user runs is they can be seen.

- Finish README.md and PATTERN.md. Explain one event, method, validation, UI, timer, media, collection or file pattern that can be adapted later.



PATTERN.md

PATTERN.md

Reusable Pattern: Validate a Selection Before Acting

Purpose

Check that the user has selected a valid item before performing an action. This prevents the app from opening a link when no link is selected.

Where It Is Used

In `MainWindow.xaml.cs`, the `OpenSelectedLink()` method checks the selected item in the `lstLinks` ListBox before opening the link.

Example

```
if (lstLinks.SelectedItem is not string selectedLink)
{
    txtStatus.Text = "Please select a link first.";
    return;
}
```

How It Works

- `lstLinks` is the ListBox containing the saved links.
- `SelectedItem` returns the currently selected item. It can be `null` when nothing is selected.
- `is not string selectedLink` checks whether the selected item is a string. If it is, the value is stored in the local variable `selectedLink`.
- `txtStatus.Text` displays a message when no valid link is selected.
- `return` exits the method immediately, preventing the remaining code from opening a link.

Related UI Event

The `Window_MouseDown` event handler clears the selected link when the user clicks outside the links ListBox.

```
private void Window_MouseDown(object sender, MouseButtonEventArgs e)
{
    if (!lstLinks.IsMouseOver)
    {
        lstLinks.SelectedItem = null;
    }
}
```

The event handler is connected in `MainWindow.xaml` using `MouseDown="Window_MouseDown"`.

This allows the selection validation to work when the user clicks elsewhere in the window.

Test

1. Add a valid URL to the list.
2. Select the URL and click Open Selected Link. Confirm that it opens in the browser.
3. Click outside the links ListBox.
4. Click Open Selected Link again.

Expected result: The browser does not open the previous URL, and the app asks the user to select a link first.

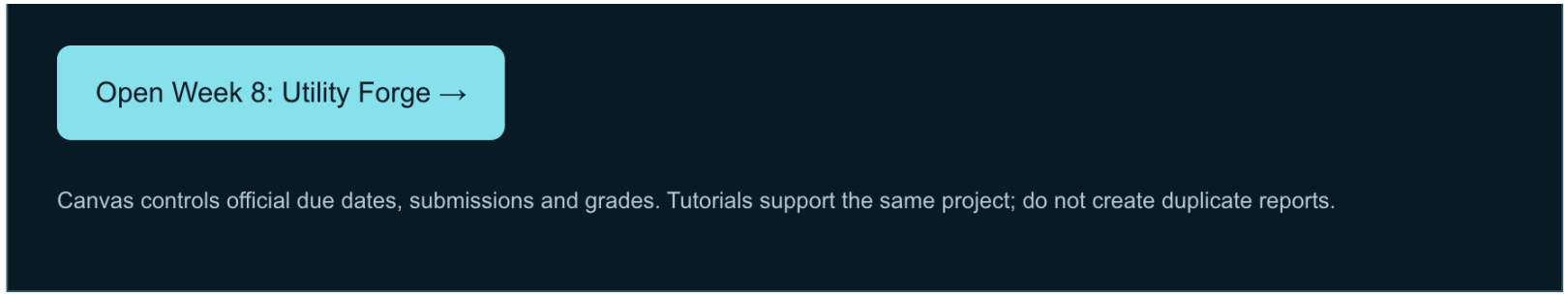
Reusing This Pattern

Use this pattern whenever an action depends on a selected item. Validate the selection before performing the action, show a clear message if the selection is missing or invalid, and use `return` to prevent the action from continuing.

This helps prevent errors and makes the application easier to use.

Submit the tested GitHub link

Link: https://github.com/saines007/CS077-Fall-2026-Team/tree/main



https://github.com/saines007/CS077-Fall-2026-Team/tree/main

Requirement 07